Portfolio Risk Analysis - One Pager

Team: Gabe Pringle, Sina Tegene, Max Fan, Zabe Nash

Date: April 30, 2025

Project Overview

We analyzed two different 2-asset portfolios to compare risk, correlation, and diversification potential:

- 60% SPY / 40% TLT (Traditional Diversified)

- 50% AAPL / 50% MSFT (High-Beta Tech Portfolio)

Portfolio Metrics Summary

Portfolio	Daily Std Dev	Portfolio Beta	Correlation (r)
SPY / TLT	0.007	0.527	-0.210
AAPL / MSFT	0.016	1.218	0.683

Visual Insights

SPY vs. TLT Residual Plot:

- Random scatter = no pattern = strong case for diversification

- Model prediction errors centered around 0 = good linear fit

AAPL vs. MSFT Residual Plot:

- Residuals tightly clustered = strong correlation

- Not ideal for diversification due to shared trends and sector exposure

Asset Snapshots - April 29, 2025

Ticker	Price	Daily Change	52W Range	Market Cap	P/E
SPY	5,560.83	+0.58%	4,835 - 6,147	-	-
TLT	$90.20	+0.85%	$84.89 - $101.64	-	-
MSFT	$394.04	+0.73%	$344 - $468	$3.13T	33.79
AAPL	$211.21	+0.51%	$169 - $260	$3.17T	40.27

Key Takeaway

While tech stocks may promise higher returns, they also carry higher risk due to correlation and beta. A portfolio blending equities with negatively correlated assets like long-term Treasuries (TLT) offers better

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downside protection and lower total volatility, making it the more resilient strategy.